Xin Zou

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Department of Mathematics

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Education

B.S. Mathematics, *University of Science and Technology of China*, 2020
Advisor: Mao Sheng

Ph.D. Mathematics, *Johns Hopkins University*, 2027
Advisor: Yiannis Sakellaridis

Research Interest

Archimedean zeta integrals , Harmonic analysis on Spherical varieties, Cohomological representation, L-functions and special L-values, Period relations.

Preprints

Archimedean Zeta Integrals for Orthogonal Gan-Gross-Prasad Pairs
Thesis, 2027.

An Arithmetic Property of Intertwining Operators for Real Reductive Groups,
joint with A. Raghuram, preprint, 2026. arXiv:2610.40261.

Invited Talks

An arithmetic property of intertwining operators for real reductive groups October 4, 2025
Invited talk at the Special Session on Automorphic Forms and Representation Theory: Current Trends, AMS Fall Southeastern Sectional Meeting.
Tulane University, New Orleans, Louisiana, USA.

Attended Conferences

Recent Progress on the Unitary Dual Problem, September 22-25, 2026
Brin Mathematics Research Center, University of Maryland, USA.

JHU–UMD Algebra and Number Theory Day November 15, 2025
Brin Mathematics Research Center, University of Maryland, USA.

JHU–UMD Algebra and Number Theory Day May 3, 2025
University of Maryland, USA.

Representation Theory and Lie Groups—In Celebration of the Career of Jeffrey Adams October 14–17, 2024
Brin Mathematics Research Center, University of Maryland, USA.

Branching Problems for Representations of Real, p-adic and Adelic Groups July 7–12, 2024
Banff International Research Station (BIRS), UBC Okanagan, Kelowna, British Columbia, Canada.

Los Angeles Workshop on Representations and Geometry (LAWRGe): Relative Langlands Duality June 10–14, 2024
University of Southern California, Los Angeles, California, USA.

JHU–UMD Algebra and Number Theory Day April 20, 2024
University of Maryland, College Park, Maryland, USA.

Special Session on Automorphic Forms and Trace Formulae: AMS Eastern Sectional Meeting. April 6–7, 2024
Howard University, Washington, DC, USA.

Johns Hopkins Junior Number Theory Days 2023/24 February 10–11, 2024
Johns Hopkins University, Baltimore, Maryland, USA.

JHU–UMD Algebra and Number Theory Day November 4, 2023
Johns Hopkins University, Baltimore, Maryland, USA.

JHU–UMD Algebra and Number Theory Day April 22, 2023
Johns Hopkins University, Baltimore, Maryland, USA.

Learning Seminars

The Fundamental Lemma Fall 2026
Yiannis Sakellaridis

Local Trace and Multiplicity Formulas for Spherical Varieties Spring 2026
Yiannis Sakellaridis

p-adic L-functions, Euler Systems and Arithmetic Applications Fall 2025
Yiannis Sakellaridis

Special Values of L-functions Spring 2025
David Savitt

Automorphic Forms Learning Seminar
Yiannis Sakellaridis

Hasse Invariants and Congruences: George Boxer's Thesis Fall 2024
David Savitt

Motives and Motivic Periods Spring 2024
David Savitt and Yiannis Sakellaridis

Automorphic Representations and Harmonic Analysis
Yiannis Sakellaridis

The Unbounded Denominators Conjecture Fall 2023
David Savitt

Automorphic Forms Learning Seminar
Yiannis Sakellaridis

Topics in Representation Theory: Companion Learning Seminar Spring 2023
Organized by Rui Chen alongside Yiannis Sakellaridis's course

Euler Systems Fall 2022
David Savitt and Yiannis Sakellaridis

Topics in Representation Theory
Yiannis Sakellaridis

Derived Galois Deformation Theory Spring 2022
David Savitt

Derived Galois Deformation Rings Fall 2021
David Savitt

Lie Groups and Quantization
Yiannis Sakellaridis

Number Theory Learning Seminar: Rigid Analytic Spaces Spring 2021
David Savitt

Teaching

Calculus III (110.202) Spring 2026
Teaching Assistant

Calculus III (110.202) Fall 2025
Teaching Assistant

Linear Algebra (110.201) Spring 2025
Teaching Assistant

Linear Algebra (110.201) Fall 2024
Teaching Assistant

Linear Algebra (110.201) Fall 2023
Head Teaching Assistant

Honors Multivariable Calculus (110.211) Spring 2023
Teaching Assistant

Elementary Number Theory (110.304) Spring 2023
Teaching Assistant

Honors Linear Algebra (110.212) Fall 2022
Teaching Assistant

Honors Complex Analysis (110.407) Fall 2022
Grader

Linear Algebra (110.201) Spring 2022
Teaching Assistant

Introduction to Abstract Algebra (110.401) Fall 2021
Teaching Assistant